

Part C

Course title: SC7 Bayes Methods
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1. Let S be a finite set and let $\mathcal{S}$ be the set of all subsets of S . Let $>, <, \sim$ be an order on sets $A, B \in \mathcal{S}$ satisfying $A \sim A$ and $A > B \Rightarrow B < A$. Here are the first three Savage Axioms.

SA1. For any $A, B \in \mathcal{S}$ exactly one of the following relations holds: $A > B$, $A < B$, $A \sim B$.

SA2. For all $A_1, A_2, B_1, B_2 \in \mathcal{S}$, if $A_1 \cap A_2 = B_1 \cap B_2 = \emptyset$ and $A_i \geq B_i$, $i = 1, 2$ then $A_1 \cup A_2 \geq B_1 \cup B_2$. If in addition either $A_1 > B_1$ or $A_2 > B_2$ then $A_1 \cup A_2 > B_1 \cup B_2$.

SA3. If $A \in \mathcal{S}$ then $\emptyset \leq A$ with $\emptyset < S$.

- (a) [10 marks] Let three sets $A, B, D \in \mathcal{S}$ be given.

(i) Show that $A \geq B$ implies $A^c \leq B^c$ where A^c is the complement of A in S .

(ii) Suppose $D \cap A = D \cap B = \emptyset$. Show that $A \geq B$ if and only if $A \cup D \geq B \cup D$.

(iii) Show that $A \subseteq B$ implies $B \geq A$.

- (b) [7 marks] Let $z = (z_1, z_2)$ with $z \in S$ and $S = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$ be a model index. Consider a linear model with covariates $x_{i,1}, x_{i,2}$, parameters $\beta_1, \beta_2 \in R$, linear predictor $\eta_i = x_{1,i}\beta_1 z_1 + x_{2,i}\beta_2 z_2$ and $y_i \sim p(\cdot | \eta_i)$ jointly independent for $i = 1, \dots, n$ given $\eta_1, \dots, \eta_n$. Let $y = (y_1, \dots, y_n)$.

Assume the marginal likelihoods $p(y|z)$, $z \in S$ are finite and let $\pi(z)$ be the prior and $\pi(z|y)$ the posterior over models $z \in S$. For $A \in \mathcal{S}$ let

$$p(y|A) = \sum_{z \in A} p(y|z) \pi(z|z \in A) \quad \text{and} \quad \pi(A|y) = \sum_{z \in A} \pi(z|y)$$

- (i) Suppose that for all $A, B \in \mathcal{S}$ our preferences satisfy $A < B \Leftrightarrow \pi(A|y) < \pi(B|y)$. Explain briefly why these preferences necessarily satisfy the Savage Axioms.

- (ii) Suppose now that our preferences over $A, B \in \mathcal{S}$ are given by Bayes factors, so $A < B \Leftrightarrow p(y|A) < p(y|B)$. Under what conditions, if any, do these preferences satisfy the Savage axioms?

- (c) [8 marks] Consider a new axiom SA2':

SA2': for $A, B, C \in \mathcal{S}$ if $A \geq B$ and $B \geq C$ then $A \geq C$.

- (i) Show that axioms SA1, SA2 and SA3 imply SA2'.

- (ii) Show that there exist preferences that satisfy axioms SA1, SA2' and SA3 but do not satisfy SA2 and so SA2 and SA2' are not equivalent.

Hint: take a small set S and specify transitive order relations $>, <, \sim$ satisfying SA1, SA2' and SA3 between all sets in $\mathcal{S}$. Design the relations so they don't satisfy some simple property that follows from SA2.

2. (a) [8 marks]

- (i) Give the definition of an infinite exchangeable sequence.
- (ii) For fixed $K \geq 1$ and $\alpha_j > 0$, $j = 1, \dots, K$ let $\alpha = (\alpha_1, \dots, \alpha_K)$. Let $\Delta_K = \{q \in [0, 1]^K : \sum_{j=1}^K q_j = 1\}$ and $q \sim f(\cdot; \alpha)$ with $f(q; \alpha)$ the Dirichlet density

$$f(q; \alpha) = \frac{\Gamma(\sum_j \alpha_j)}{\prod_j \Gamma(\alpha_j)} \prod_{j=1}^K q_j^{\alpha_j-1}, \quad q \in \Delta_K.$$

For $i \geq 1$ let $Z_i \sim \text{Multinom}(1, q)$ be an infinite sequence of multinomial random variables which are jointly independent given q , so $p(Z_i = j|q) = q_j$ for $j = 1, \dots, K$ and $i \geq 1$. For $z_i \in \{1, \dots, K\}$ let

$$p(z_1, \dots, z_n) = \Pr(Z_1 = z_1, \dots, Z_n = z_n).$$

Let $n_j = \sum_{i=1}^n \mathbb{I}_{z_i=j}$, $j = 1, \dots, K$.

Calculate $p(z_1, \dots, z_n)$ as an explicit function of n, K, α_j and n_j , $j = 1, \dots, K$ and show that Z_i , $i \geq 1$ is an infinite exchangeable sequence.

- (b) [9 marks] Let $x_i \in \{1, \dots, K\}$, $i \geq 1$ be categorical variables with K levels and let $\beta = (\beta_1, \dots, \beta_K)$, $\beta \in R^K$ be a parameter vector. Consider the normal linear model $Y_i = \beta_{x_i} + \epsilon_i$ with $\epsilon_i \sim N(0, \sigma^2)$ jointly independent for $i \geq 1$.
- (i) For $j \in \{1, \dots, K\}$, let $n_j^{(i)} = \sum_{a=1}^i \mathbb{I}_{x_a=j}$ give the number of times level j appears in $x_{1:i} = (x_1, \dots, x_i)$. The x_i -values are a realisation of a sequence of random variables X_i , $i \geq 1$ with $X_1 \sim U\{1, \dots, K\}$ and for $i \geq 1$,

$$\Pr(X_{i+1} = j | X_1 = x_1, \dots, X_i = x_i) = \frac{n_j^{(i)} + 1}{i + K}.$$

Calculate $\pi_X(x_{1:n}) = \Pr(X_1 = x_1, \dots, X_n = x_n)$ and show that $X_1, X_2, X_3, \dots$ is an infinite exchangeable sequence.

- (ii) A prior $\pi_B(\beta)$ for $\beta \in R^K$ is given. Let

$$p_{Y|X}(y_1, \dots, y_n | x_1, \dots, x_n) = \int_{R^K} p_{Y|X,\beta}(y_1, \dots, y_n | x_1, \dots, x_n, \beta) \pi_B(\beta) d\beta.$$

Show that, for $\sigma = (\sigma_1, \dots, \sigma_n)$ any permutation of $\{1, 2, \dots, n\}$,

$$p_{Y|X}(y_1, \dots, y_n | x_1, \dots, x_n) = p_{Y|X}(y_{\sigma_1}, \dots, y_{\sigma_n} | x_{\sigma_1}, \dots, x_{\sigma_n})$$

and hence show $(X_1, Y_1), (X_2, Y_2), (X_3, Y_3), \dots$ is an infinite exchangeable sequence.

(c) [8 marks]

- (i) Write down the posterior $\pi(x_{1:n}, \beta | y_{1:n})$ in terms of the model elements.
- (ii) Give a Gibbs sampler for $x_1, \dots, x_n$ given $y_{1:n}$ and β .

3. (a) [9 marks]

- (i) Let $u \sim g(u)$, $u \in \mathcal{U}$ be a continuous random variable with density g on $\mathcal{U} = \mathbb{R}^d$. For $\theta \in \Omega$ with $\Omega = \mathbb{R}^p$ and $p \geq d$, let $\theta' = \psi_1(\theta, u)$ where ψ_1 is invertible in u and θ' at fixed θ . As a further condition on ψ_1 , the reverse update $\theta = \psi_1(\theta', u')$ has a unique solution in u' at fixed (θ, θ') . Denote by $u' = \psi_2(\theta, u)$ the solution to $\theta = \psi_1(\psi_2(\theta, u), u')$.

Let $\psi(\theta, u) = (\psi_1(\theta, u), \psi_2(\theta, u))$. Show that $\psi(\psi(\theta, u)) = (\theta, u)$.

- (ii) Assume ψ is differentiable in both arguments. The following MCMC update targets a density $\pi(\theta)$, $\theta \in \Omega$. Let $X_t = \theta$ with $\pi(\theta) > 0$.

(1) Simulate $u \sim g(u)$ and set $(\theta', u') = \psi(\theta, u)$. Let

$$J_\psi(\theta, u) = \left| \frac{\partial(\theta', u')}{\partial(\theta, u)} \right|$$

be the Jacobian for the transformation $(\theta', u') = \psi(\theta, u)$.

(2) With probability

$$\alpha(\theta'|\theta) = \min \left\{ 1, \frac{\pi(\theta')g(u')}{\pi(\theta)g(u)} J_\psi(\theta, u) \right\}$$

set $X_{t+1} = \theta'$ and otherwise set $X_{t+1} = \theta$.

Show that this update satisfies detailed balance

$$\pi(\theta)g(u)\alpha(\theta'|\theta)d\theta du = \pi(\theta')g(u')\alpha(\theta|\theta')d\theta' du'$$

with respect to $\pi(\theta)$. *Hint: you may assume $J_{\psi^{-1}}(\theta', u') = J_\psi(\theta, u)^{-1}$.*

- (b) [8 marks] Let $\pi(\theta|y) \propto \pi(\theta)p(y|\theta)$, $\theta \in \mathbb{R}^2$ be a posterior for a given observation model $p(y|\theta)$ and prior $\pi(\theta) = N(\theta; \underline{0}, I_2)$ with $\underline{0} = (0, 0)^T$.

- (i) Let $0 < \delta \leq \pi$ (radians) be given. For $\theta \in \mathbb{R}^2$ and $u \in [-\delta, \delta)$ let $\theta' = M(u)\theta$ where

$$M(u) = \begin{pmatrix} \cos(u) & -\sin(u) \\ \sin(u) & \cos(u) \end{pmatrix}$$

Here is an MCMC update targeting $\pi(\theta|y)$. Let $X_t = \theta$ with $\pi(\theta|y) > 0$.

1. Simulate $u \sim U(-\delta, \delta)$ and set $\theta' = M(u)\theta$.
2. With probability $\alpha(\theta'|\theta)$ set $X_{t+1} = \theta'$ and otherwise set $X_{t+1} = \theta$.

Calculate the acceptance probability $\alpha(\theta'|\theta)$ simplifying as far as you can.

- (ii) Let $v \sim U(w, 1/w)$ with $0 < w < 1$ and v independent of u . Replace the proposal $\theta' = M(u)\theta$ with $\theta' = vM(u)\theta$. Calculate the new acceptance probability.

- (c) [8 marks] Consider two observation models $p_1(y|\phi)$ and $p_2(y|\psi)$ with $\phi \in \mathbb{R}^2$ and $\psi \in \mathbb{R}^3$ and prior densities $\pi_1(\phi)$ and $\pi_2(\psi)$. The model-prior is $\pi_M(m) = 1/2$, $m \in \{1, 2\}$.

- (i) Give the joint posterior $\pi(\cdot, m|y)$ in terms of the model elements.

- (ii) Consider the following MCMC update. Let $X_t = (\theta, m)$ with $\pi(\theta, m|y) > 0$.

1. If $m = 1$ simulate $\theta' \sim \pi_2(\cdot)$ and set $m' = 2$, otherwise $\theta' \sim \pi_1(\cdot)$ and $m' = 1$.
2. With probability

$$\alpha(\theta', m'|\theta, m) = \min \left\{ 1, \frac{p_{m'}(y|\theta')}{p_m(y|\theta)} \right\}$$

set $X_{t+1} = (\theta', m')$ and otherwise set $X_{t+1} = (\theta, m)$.

Write down detailed balance for this update for two states $(\theta, m) = (\phi, 1)$ and $(\theta', m') = (\psi, 2)$ and verify it holds.